

LCAnet: A Modular Ontology Network for Life Cycle Assessment Knowledge Modeling

Fatima Danash^{1,*}, Chokri Mraidha¹

¹Université Paris-Saclay, CEA, List, F-91120, Palaiseau, France

Abstract

Life Cycle Assessment (LCA) is a standardized methodology for evaluating the environmental impacts of products across their life cycle. Despite its widespread adoption, LCA knowledge remains fragmented across incompatible tools and data formats, hindering semantic interoperability and reuse. Existing ontological approaches address isolated aspects of LCA while none provides a unified, semantically rich framework covering the full spectrum of LCA modeling. We present LCAnet, a modular OWL ontology network structured into six interoperable modules: Core, Activity, Actor, Property, Spatiotemporal, and Impact Assessment. To the best of our knowledge, no prior LCA ontology simultaneously covers all principal aspects of an LCA study within a single, cohesive, and publicly available framework. Beyond taxonomic coverage, LCAnet draws on established ontology design pattern principles and provides rich formal semantics that capture domain constraints and enable automated inference. Applicability is demonstrated on an integrated circuit manufacturing scenario, with additional instantiations in battery and additive manufacturing domains. We evaluate LCAnet against 20 competency questions verified through SPARQL queries over a populated knowledge graph, and validate logical consistency using HermiT and Pellet. External reuse is evidenced by the independent adoption of LCAnet’s Impact Assessment module within an EU project for Digital Product Passport ontology development. The ontology is documented and publicly available with persistent URIs under w3id.org/lcanet.

Keywords

Life Cycle Assessment, Modular Ontology, Semantic Interoperability, Ontology Network

1. Introduction

Life Cycle Assessment (LCA) is a widely adopted methodology for evaluating the environmental impacts of products and systems across their full life cycle [1]. Despite its standardisation and broad industrial uptake, LCA knowledge remains fragmented across incompatible tools, proprietary data formats, and isolated databases—severely hindering semantic interoperability, data reuse, and cross-system integration. The heterogeneous nature of LCA data, spanning activities, flows, actors, spatial and temporal contexts, quantitative and qualitative properties, impact characterizations, and study metadata, makes this fragmentation particularly pronounced: no shared, machine-interpretable representation exists that covers the full spectrum of LCA modeling.

Ontologies offer a principled solution to this challenge, enabling formal knowledge representation, semantic interoperability, and automated reasoning over heterogeneous data sources. Several ontology-based approaches have been proposed for LCA over the past decade; however, as surveyed in Section 2, none provides a unified, semantically rich, and comprehensive framework covering all aspects of LCA. Three gaps are particularly evident: impact assessment remains unaddressed by all existing LCA ontologies; expressive axiomatization and rule-based inference are consistently absent; and modular architecture enabling independent reuse of individual components has not been adopted in prior work.

The main contribution of this paper is **LCAnet**, a modular OWL 2 DL ontology network for LCA designed to address the literature’s gaps and provide a unified semantic framework for LCA knowledge modeling. The design of LCAnet draws on established ontology design patterns, where each ontology module behaves as an ontology design pattern, which enables the independent reuse of domain-specific

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*Corresponding author.

✉ fatme.danash@cea.fr (F. Danash); chokri.mraidha@cea.fr (C. Mraidha)

ORCID: 0000-0002-0438-6831 (F. Danash); 0000-0003-2993-5734 (C. Mraidha)



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components. To the best of our knowledge, LCAnet is the first LCA ontology to simultaneously address all these dimensions within a single, cohesive, and publicly available framework. Beyond taxonomic structure, LCAnet achieves semantic richness through expressive OWL axiomatization and SWRL inference rules that capture domain constraints and enable automated reasoning. Applicability is evaluated through 20 competency questions verified by SPARQL queries over a populated knowledge graph of 278 individuals instantiating the StorAlge integrated circuit prototype, with logical consistency confirmed using HermiT and Pellet. Evidence of reuse is provided through an additional instantiation for a battery manufacturing use case and the independent adoption of LCAnet’s Impact Assessment module within the Cirpass-2 EU project for Digital Product Passport ontology development. The ontology is documented and publicly available with persistent URIs under <https://w3id.org/lcanet>.

The remainder of this paper is structured as follows. Section 2 provides background on LCA and reviews existing ontological approaches. Section 3 presents the design requirements and choices that motivated LCAnet’s architecture. Section 4 describes the LCAnet ontology network design and implementation. Section 5 instantiates LCAnet on the StorAlge prototype use case. Section 6 evaluates LCAnet across quality, reusability, and impact dimensions. Section 7 concludes the paper.

2. Background Related Work

LCA is a standardized methodology for evaluating the environmental impacts of products, processes, or systems across their entire life cycle—from raw material extraction to end-of-life disposal [1]. It supports comprehensive sustainability assessments by accounting for multiple impact categories and life cycle stages [2], and informs eco-design, regulatory decisions, and sustainability strategies across industries [3]. An LCA study is structured into four phases: goal and scope definition, life cycle inventory (LCI) analysis, life cycle impact assessment (LCIA), and interpretation. Each phase involves heterogeneous data presenting different aspects of LCA—activities, flows, actors, spatial and temporal contexts, properties (qualitative and quantitative), impact characterizations, and the LCA study’s metadata—whose formal, machine-interpretable representation remains a significant open challenge.

Ontologies—formal, explicit specifications of shared conceptualizations [4, 5]—offer a principled solution to the semantic fragmentation of LCA data, enabling structured knowledge representation, interoperability, and reasoning across heterogeneous sources [6]. Several ontology-based approaches have been proposed for LCA, though none has yet achieved comprehensive coverage or wide adoption.

The earliest dedicated effort was the Earthster Core Ontology (ECO), developed as part of the Earthster open-source initiative for LCA linked data applications [7, 8]. Despite its early promise, ECO received limited community engagement and development stopped in 2011. Around the same period, a number of independent and fragmented initiatives emerged, targeting specific industrial sectors such as refinery operations and product manufacturing [9, 10, 11], but these lacked the formal ontological structure necessary for semantic interoperability [12].

A more principled effort was introduced by Janowicz et al. with the LCA Ontology Design Pattern (LCA-ODP) [13], which provided a formal description logic model for LCA activities and flows, with partial coverage of actors (Agent) and a single generic property class (Property). Concurrently, Yan et al. proposed STScopes [14], a follow-up of LCA-ODP specifically addressing the spatiotemporal aspect of life cycle inventory data through dedicated spatiotemporal classes (SpatiotemporalRequirement, SpatiotemporalScope, and SpatiotemporalOutcome). Both ontologies were specified in description logics and later implemented in OWL; however, their source files have not been made publicly available and development ceased after 2015. These ontologies served as the semantic foundation for generating LCA catalogs from heterogeneous LCI datasets such as Ecoinvent v3.2 and GaBi pro (the 2016 edition with 22 extensions databases) in [15]—a data engineering effort rather than an ontology contribution.

Building on this foundation, Ghose et al. introduced the BONSAI ontology (Big Open Network for Sustainability Assessment Information) [16], extending both LCA-ODP and STScopes to support broader integration of Life Cycle Sustainability Assessment (LCSA) data. BONSAI defines core classes for activities, flows, and agents, reuses external vocabularies for measurement units (Unit, Measure,

Table 1

Comparison of existing approaches across key LCA modeling components. ✓ = covered, ∂ = partially covered, ✗ = not covered.

Aspect	ECO (2010)	LCA-ODP (2015)	STScopes (2015)	BONSAI (2019)	CEON (2023)
Activity/Flow	✓	✓	✓	✓	∂
Actors	∂	∂	✗	✓	✓
Properties	∂	∂	✗	∂	∂
Spatiotemporal	✗	✓	✓	∂	✓
Impact Assessment	✗	✗	✗	✗	✗
Semantic richness	✗	∂	∂	∂	✓
Modular	✗	✗	✗	✗	✓
Public & active	✗	✗	✗	∂	✓

Quantity, and NumericalValue from the Ontology of units of Measure OM), and remains the most comprehensive publicly available LCA ontology to date, with its OWL implementation last updated in 2022 [17]. This work served as a basis for RDF conversion workflows of datasets such as Exiobase [18] into RDF graphs and input-output datasets [19], another data engineering contribution. The associated infrastructure continues to support the community’s transition toward FAIR data principles [20]. However, BONSAI does not address impact assessment, its property model is limited to a single balanceable property type linked to the OM units ontology, its spatiotemporal coverage reduces to a location class reused from schema.org, and its semantic richness in terms of OWL axiomatization and rule-based inference remains limited.

Beyond the LCA-specific literature, the Circular Economy Ontology Network (CEON) [21] represents a structurally relevant parallel effort. CEON is a modular OWL ontology network for circular economy modeling, covering processes, resources, and actors through reusable ontology design patterns, with a spatial coverage via GeoSPARQL alignment. While CEON covers several key modeling components, its abstractions are primarily tailored to circular economy use cases. Processes are modeled with inputs and outputs (not necessarily resources), linked to locations, and structured along product value chains—design choices well suited to CE objectives. However, these do not fully capture LCA-specific requirements, such as the explicit treatment of elementary flows, technosphere–biosphere distinctions, impact assessment modeling, and the LCA study metadata.

Table 1 summarizes the coverage of existing approaches across the principal aspects of LCA modeling. These aspects encompass: core LCA components (activities/flows and actors), contextual dimensions (spatial and temporal), and descriptive layers (properties, impact characterization, and metadata). As the table illustrates, existing approaches collectively address several dimensions but none simultaneously covers all principal aspects within a single, cohesive, and publicly available framework. Three gaps are particularly evident. First, impact assessment—a core phase of any LCA study—remains unaddressed by all surveyed ontologies. Second, semantic richness beyond basic taxonomic structure is consistently absent: no existing LCA ontology (excluding CEON as it is not an LCA ontology) provides expressive OWL axiomatization combined with rule-based inference. Third, with the exception of CEON, none of the surveyed ontologies adopts a modular architecture, limiting the independent reuse of individual components. These gaps directly motivate the design of LCA_{net}, as elaborated in Section 3.

3. Requirements and Design Considerations

The requirements presented in this section were identified through a series of working sessions conducted with LCA domain experts within the SmartACV Carnot project, an ANR-funded French research initiative. These sessions informed both the conceptualization of the domain and the formulation of the competency questions used to evaluate LCA_{net}.

3.1. Design Requirements

Based on the LCA modeling challenges and the gaps identified in Section 2, a set of design requirements (DRs) was defined to guide the development of LCAnet. These requirements reflect the need to represent the multi-aspect nature of LCA, which spans several interconnected subdomains, including process modeling, organizational modeling, quantitative analysis, spatiotemporal representation, and impact assessment.

- DR1** Represent the core structure of an LCA study, including goal, scope, system boundaries, and functional unit.
- DR2** Model LCA activities and flows, including inputs, outputs, and exchanges between processes.
- DR3** Represent actors and their roles within LCA systems, including responsibilities and participation contexts.
- DR4** Represent quantitative and qualitative properties associated with flows, activities, and results.
- DR5** Capture the spatiotemporal context of activities and inventory data.
- DR6** Model the impact assessment of activities and flows, including methods, impact categories, indicators, and characterization factors.
- DR7** Enable automated inference through formal axioms and rules to support knowledge discovery, classification, and validation.
- DR8** Support modular reuse, allowing each part of the ontology to be used independently or combined as needed.
- DR9** Ensure interoperability through the reuse and alignment with established ontologies and vocabularies.

3.2. Competency Questions

To validate the coverage and applicability of LCAnet, a set of competency questions (CQs) was defined by LCA experts, following established ontology engineering practices. These questions reflect typical queries posed by LCA practitioners and are used to assess whether the ontology can adequately represent and retrieve relevant knowledge.

The CQs defined span multiple aspects of LCA, including activity modeling, flow characterization, LCA system modeling, impact assessment, respective actors, spatiotemporal context, and descriptive properties. Each CQ is associated with one or more design requirements, ensuring traceability between requirements and their coverage. Table 2 presents 10 representative subset of the CQs used in this work.

3.3. Design Choices

To address the design requirements outlined in Section 3.1, LCAnet adopts a modular ontology architecture, in contrast to monolithic approaches commonly found in existing LCA ontologies. While monolithic ontologies may simplify initial development, they tend to obscure the intrinsic multi-aspect of the LCA domain. The decision to adopt a modular architecture is motivated by (a) the need to explicitly capture these distinct yet interrelated concerns. Each module is designed to address a specific subset of the domain, thereby supporting [DR1–DR6] through a clear separation of concerns. This design enables a more faithful representation of LCA knowledge while maintaining conceptual clarity. Modularity further (b) supports reuse and adoption flexibility [DR8]. Individual modules can be reused independently in different contexts without requiring the full ontology network. Such a design choice aligns with established ODP practices that promote modularity, reusability, maintainability, and extensibility within ontology networks. In addition, modularization (c) facilitates maintainability

Table 2

Representative list of competency questions for LCAnet

ID	Competency Question	DRs
CQ1	What are the input and output flows of all activities?	DR2
CQ2	What is the dependency chain of activities in a system model?	DR2
CQ3	Which flow datasets come from database X?	DR2, DR4
CQ4	What are the elementary and intermediate flows associated with a given activity?	DR2, DR7
CQ5	Which flows are classified as resources, emissions, or products?	DR7
CQ6	What is the functional unit, scope, system boundary, and responsible agent of the LCA model?	DR1, DR3, DR5
CQ7	What is the impact result of a given activity or flow across impact categories?	DR6
CQ8	Which flows contribute significantly (e.g., > 2%) to a given impact category?	DR6, DR7
CQ9	What are the top contributors to an impact category for a given activity?	DR6
CQ10	What is the total amount of flow X generated within the system boundary?	DR2, DR4

and extensibility. Each module can evolve independently, allowing for the integration of alternative modeling choices or alignment with external ontologies without affecting the entire network. This is particularly important in LCA, where methodologies, impact categories, and data models are subject to continuous evolution.

To ensure interoperability (DR9), LCAnet reuses and aligns with established ontologies and vocabularies, including QUDT¹ for quantities and units, OWL-Time² for temporal concepts, FOAF³ and Schema.org⁴ for agents and location, GeoNames⁵ for geographic entities, and Dublin Core Terms⁶ for metadata. This reuse promotes semantic compatibility with existing Linked Data resources and reduces modeling redundancy. Entity-level reuse of BONSAI was not pursued due to its limited maintenance and lack of impact assessment coverage. CEON’s abstractions target circular economy use cases and are not directly compatible with LCA-specific flow typing requirements.

Finally, LCAnet incorporates formal axioms and SWRL rules to support automated reasoning (DR7). SWRL is used where OWL 2 DL expressivity is insufficient, specifically for conditional multi-body inferences such as flow propagation across composed activities and impact category inheritance across aggregation levels.

4. LCAnet: Ontology Network Design and Implementation

4.1. Overview of LCAnet

The LCAnet ontology network is a modular ontology composed of six ontology modules: Activity, Actor, Property, Spatiotemporal, Impact Assessment, and Core. Together, these modules enable the semantic representation of LCA knowledge across multiple complementary dimensions. The design of LCAnet builds on several established ontology design patterns (ODPs). The Property module adopts the *n-ary relation pattern* [22] to represent dynamic and heterogeneous attributes without modifying the ontology schema. The modular architecture itself instantiates the *modular ontology pattern*, enabling separation of concerns and partial adoption. The Actor module applies the *agent role pattern*, reifying roles and responsibilities as first-class entities rather than encoding them as subclasses of Agent. These design choices reflect established ODP practice and contribute to the reusability and extensibility of the ontology network.

¹<http://qudt.org/schema/qudt#>

²<http://www.w3.org/2006/time#>

³<http://xmlns.com/foaf/0.1/>

⁴<https://schema.org/>

⁵<http://www.geonames.org/ontology#>

⁶<http://purl.org/dc/terms/>

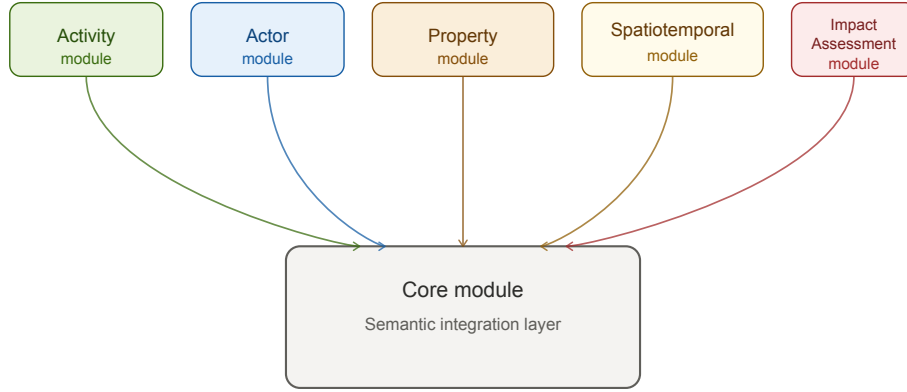


Figure 1: Architecture of the LCAnet ontology network and relations between its ontology modules.

Each module addresses a specific aspect of the LCA domain, thereby supporting a clear separation of concerns. The Activity module focuses on processes and flows, the Actor module represents organizational entities and responsibilities, the Property module captures quantitative and qualitative properties, the Spatiotemporal module models spatial and temporal contextualization, and the Impact Assessment module represents impact assessment methods and results. The Core module serves as the semantic integration layer of the ontology network. It imports the other ontology modules and defines the cross-module relations and rules required to interconnect them consistently. In addition, the core module also defines descriptive metadata elements concerning of LCA study. Consequently, individual modules remain self-contained and reusable independently, while semantic relations spanning multiple modules are centralized within the Core module. This design avoids unnecessary interdependencies between modules and facilitates extensibility and maintenance.

Figure 1 illustrates the overall architecture of the LCAnet ontology network and the relations between its constituent modules.

4.2. Ontology modules

The following subsections briefly describe the content and role of each ontology module.

4.2.1. Activity Module

The Activity module represents LCA activities and their associated flows, enabling the semantic modeling of activity/flow diagrams across the life cycle of a product or service, with central concepts such as Activity and Flow. A Flow represents a material or energy exchange associated with an activity, either as an input or output. It is specialised into types reflecting standard LCA practice: elementary, intermediate, product, waste, emission, and resource flows. The module further introduces the `ComposedActivity` class to represent activities composed of sub-activities, with accompanying SWRL rules that propagate flow relations across compositional boundaries. For instance, the following rule ensures that if a flow is an input to a sub-activity of a composed activity, it is inferred to be an input to the composed activity as well:

```

ComposedActivity(?a), Flow(?f), hasActivityPart(?a,?b),
isInputProductOf(?f,?b), Activity(?b)
→ isInputProductOf(?f,?a)

```

This inference capability supports consistent reasoning over nested activity structures, which are common in industrial LCA models.

4.2.2. Actor module

The Actor module represents the agents involved in LCA processes, together with their roles and responsibilities. Its central class `Agent` is declared equivalent to `foaf:Agent` and `dcterms:Agent`, ensuring alignment with established Semantic Web vocabularies. Specializations cover `Person`, `Group`, and `Organization`. The module introduces `Role` and `Responsibility` as first-class concepts, linked via `playsRole` and `bearsResponsibility`, enabling the contextualised assignment of roles and responsibilities to agents. A `Responsibility` is not assigned directly but is entailed by a `Role` and propagated to agents via a SWRL rule. The propagation of a responsibility across roles is formalised through the following SWRL rule:

```
Agent(?a), playsRole(?a,?r), Role(?r),  
entails(?r,?x), Responsibility(?x)  
→ bearsResponsibility(?a,?x)
```

This rule automates the inference of responsibilities from role assignments, reducing the modelling burden on data producers and ensuring consistency across actor descriptions.

4.2.3. Property module

The Property module provides a flexible representation of the qualitative and quantitative properties of LCA entities. Rather than a fixed set of datatype properties, the module adopts a *property-as-individual* pattern based on an n-ary relation modeling approach [22, 23]. A general `Property` class — with subclasses `QuantitativeProperty`, `QualitativeProperty`, and `BalanceableProperty` — is linked to subject entities via `hasProperty`. Each `Property` instance encapsulates its value and, where applicable, a unit of measure via alignment with QUDT. This design supports extensibility — novel properties can be instantiated without schema modification — and flexibility, accommodating both qualitative and quantitative attributes alongside provenance metadata.

4.2.4. Impact Assessment module

The Impact Assessment (IA) module represents impact assessment concepts, methods, and results. Its principal classes — `ImpactCategory`, `Methodology`, `ImpactCategoryIndicator`, `LCAResult`, and `CharacterizationFactor` — collectively support the representation of assessment methodologies (e.g., ReCiPe, CML), their associated impact categories, and the quantitative characterisation factors linking inventory flows to impact scores. SWRL rules are employed to propagate impact category associations across result aggregation boundaries. For example, the following rule infers that an overall impact result inherits the impact categories of the individual results it aggregates:

```
hasImpactCategory(?a,?c), ImpactCategory(?c),  
containedInOIR(?a,?o), LCAResult(?a),  
OverallImpactResult(?o)  
→ hasImpactCategory(?o,?c)
```

4.2.5. Spatiotemporal module

The Spatiotemporal (ST) module represents the spatial and temporal context of LCA processes. For temporal representation, it imports the OWL Time ontology [24], reusing `time:Instant`, `time:Interval`, and `time:DurationDescription`. For spatial representation, it defines a `Location` class aligned with `gn:Feature` (GeoNames) and `schema:Place`, with subclasses `Site` and `AdministrativeRegion`. A `SpatialScale` class with subclasses `Local`, `National`, `Regional`, and `Global` enables explicit specification of spatial granularity — a dimension relevant to both inventory modeling and impact characterisation. Reuse of established vocabularies maximises interoperability with existing geospatial and temporal knowledge graphs.

4.2.6. Core module

The Core module serves as the semantic backbone of LCAnet. It defines the foundational elements of an LCA study as specified in ISO 14040 and ISO 14044 [1], including Goal, Scope, LCAModel, LifeCycleStage, and Interpretation. Beyond its descriptive role, the Core module imports all other LCAnet modules and defines the cross-module object properties required to interconnect them consistently. Representative cross-module properties include `activityBelongsToLifeCycleStage`, linking activities to their life cycle stage; `flowHasImpactResult` and `activityHasImpactResult`, linking inventory elements to their assessed environmental impacts; `lcaModelHasAuthor`, linking an LCA model to its responsible agent; in addition to `scopeHasTemporalScale`, `scopeHasSpatialScale`, and `scopeHasReferenceFlow`, which ground the study scope in its spatiotemporal and functional context. By centralising integration logic within a single module, the architecture ensures that individual modules remain self-contained and independently reusable, while cross-domain relationships are governed by a dedicated semantic layer.

4.3. Implementation and Availability

LCAnet is implemented in OWL 2 DL, with SWRL rules extending the expressivity of the formal axiomatization beyond the OWL 2 DL fragment. The ontology was developed using the Protégé ontology editor [25]. Logical consistency was verified using both the HermiT [26] and Pellet [27] reasoners; no inconsistencies or profile violations were detected.

Namespace and URIs. Each module is assigned a dedicated persistent URI under `w3id.org/lcanet`, following the pattern `https://w3id.org/lcanet/{module}`, where `{module}` is one of: `core`, `activity`, `actor`, `property`, `st`, or `ia`. The network landing page is accessible at `https://w3id.org/lcanet`. Each module declares both an ontology IRI and a version IRI following semantic versioning (currently v2.2.0), ensuring that consumers can reference specific versions while the base IRI always resolves to the current stable release.

Documentation. Each module is documented using WIDOCO [28], which generates enriched HTML documentation automatically from the ontology annotations, including human-readable descriptions of all classes, properties, and axioms. The documentation is accessible directly via each module URI. The ontology is published under a Creative Commons Attribution 4.0 International licence (CC BY 4.0), permitting unrestricted reuse with attribution.

5. Use case: StorAlge prototype

5.1. Scenario Description

To demonstrate the applicability of LCAnet, we instantiate the ontology network on a real-world LCA study conducted within the SmartACV project called the StorAlge study. This study is aimed at evaluating the environmental impact of a Near Memory Computing prototype based on C-SRAM technology.

The LCA study was conducted using the EIME software tool⁷, connected to a Bill of Materials describing the electronic components of the prototype. Inventory data were sourced from two databases: CODDE (v3.00.000) and NegaOctet. Environmental impact calculations were performed using the Environmental Footprint 3.0 (EF 3.0) methodology. The resulting inventory and impact data, together with study metadata including hypotheses, parametrisation, system boundary, and actor information, were extracted and subsequently used as input data for the use case⁸.

⁷<https://ghgprotocol.org/eime-environmental-improvement-made-easy>

⁸<https://danashfatima.github.io/lcanet/ICdata/inputs/>

5.2. Knowledge Graph Instantiation

The StorAlge LCA study is represented as a populated LCAnet knowledge graph⁹, comprising 278 individuals, instantiating classes across all six ontology modules. The Core module captures the structure of the LCA study itself, including the `LcaModel`, `Goal`, `Scope`. The Activity module represents the activity/flow diagram of the prototype manufacturing process. The Actor module describes the persons involved in the study. The Property module captures quantitative and qualitative properties of flows and activities, including database provenance metadata (e.g., CODDE, NegaOctet) represented as `QualitativeProperty` individuals. The Spatiotemporal module grounds the study in its geographic and temporal context, instantiating `Location`, and `SpatialScale` individuals alongside temporal intervals reused from the OWL Time ontology. The Impact Assessment module represents the EF 3.0 impact categories, characterisation factors, and both activity-level and overall impact results.

5.3. Querying the Knowledge Graph

We demonstrate the querying capability of LCAnet through three representative SPARQL queries, selected to illustrate single-module retrieval, cross-module integration, and inference-dependent querying respectively. The full set of queries corresponding to all CQs identified in Section 3.2 is available in the LCAnet GitHub repository¹⁰.

Query 1 – LCA study metadata (CQ6). The following query retrieves the complete metadata of the LCA study, spanning goal, life cycle stages, functional unit, reference flow, system boundary, spatiotemporal scales, and assumptions, demonstrating the descriptive coverage of the Core module:

Listing 1: Retrieving LCA study metadata (CQ6).

```
PREFIX core: <https://w3id.org/lcanet/core#>

SELECT ?lcaModel
  (SAMPLE(?goal) AS ?goal)
  (GROUP_CONCAT(DISTINCT ?lcStage; separator=",_") AS ?lcStages)
  (SAMPLE(?functionalUnit) AS ?functionalUnit)
  (SAMPLE(?referenceFlow) AS ?referenceFlow)
  (GROUP_CONCAT(DISTINCT ?ass; separator="_|_") AS ?assumptions)
  (SAMPLE(?systemBoundary) AS ?systemBoundary)
  (SAMPLE(?temporalScale) AS ?temporalScale)
  (SAMPLE(?spatialScale) AS ?spatialScale)
WHERE {
  ?lcaModel a core:LcaModel ;
    core:lcaModelConcernsLcStage ?lcStage ;
    core:lcaModelHasScope ?scope .
  ?scope core:scopeHasFunctionalUnit ?functionalUnit ;
    core:scopeHasReferenceFlow ?referenceFlow .
  OPTIONAL {
    ?scope core:scopeHasTemporalScale ?temporalScale ;
      core:scopeHasSpatialScale ?spatialScale ;
      core:scopeHasSystemBoundary ?systemBoundary ;
      core:scopeHasAssumption ?ass .
  }
  OPTIONAL { ?lcaModel core:lcaModelHasGoal ?goal . }
}
GROUP BY ?lcaModel
```

⁹<https://w3id.org/lcanet/ICdata>

¹⁰<https://danashfatima.github.io/lcanet/CQs>

Query 2 – Activity dependency chain (CQ2). The following query retrieves the recursive dependency chain of activities in the system model, traversing the flow connectivity graph using a SPARQL property path. This query demonstrates LCAnet’s ability to represent and navigate complex activity/flow diagrams – a structural characteristic of industrial LCA models that is difficult to query over flat or ad-hoc data representations:

Listing 2: Retrieving the recursive activity dependency chain (CQ2).

```
PREFIX act: <https://w3id.org/lcanet/activity#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?activity ?dependentActivity
WHERE {
  ?activity a act:Activity ;
    rdfs:label "Production_of_Mother_Board" ;
    act:hasInputProduct ?inputFlow .
  ?inputFlow act:connected ?outputFlow .
  ?outputFlow act:isOutputProductOf ?intermediateActivity .
  ?intermediateActivity
    (act:hasInputProduct /
     act:connected /
     act:isOutputProductOf)* ?dependentActivity .
}
```

Query 3 – Impact contributors above threshold (CQ8). The following query identifies all activities contributing more than 2% to the *Climate change, total* impact category, computing each activity’s relative contribution against the overall impact result. This query spans three modules – Activity, Impact Assessment, and Core – and relies on the cross-module object property defined in the Core module, `impactResultConcernsActivity`, and on the `containedInOIR` relation whose instances are partially inferred via SWRL rules. It illustrates the analytical queries that LCAnet enables over integrated LCA knowledge graphs, and which would require substantial manual data joins in the absence of a shared semantic framework:

Listing 3: Identifying activities contributing more than 2% to an impact category (CQ8).

```
PREFIX core: <https://w3id.org/lcanet/core#>
PREFIX ia: <https://w3id.org/lcanet/ia#>
PREFIX qudt: <http://qudt.org/schema/qudt/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?activity ?impactResultValue ?percentage
WHERE {
  ?impactCategory a ia:ImpactCategory ;
    rdfs:label "Climate_change,_total" .
  ?overallIIR a ia:OverallImpactResult ;
    ia:correspondsToIc ?impactCategory ;
    qudt:numericValue ?overallImpactValue .
  ?impactResult a ia:ImpactResult ;
    core:impactResultConcernsActivity ?activity ;
    ia:correspondsToIc ?impactCategory ;
    qudt:numericValue ?impactResultValue ;
    ia:containedInOIR ?overallIIR .
  BIND(?impactResultValue / ?overallImpactValue AS ?percentage)
  FILTER(?percentage > 0.02)
  FILTER(?overallImpactValue != 0)
}
ORDER BY DESC(?percentage)
```

6. Evaluation: Quality, Reusability, and Impact of LCAnet

6.1. Ontology Quality and Technical Validation

The technical correctness of LCAnet was assessed through automated reasoning, CQ validation, and ontology quality inspection. First, logical consistency was verified using the HermiT and Pellet reasoners. No inconsistencies were detected across the six ontology modules (Core, Activity, Actor, Property, Spatiotemporal, and Impact Assessment), as mentioned in Section 4.3, confirming that the class hierarchy, property restrictions, and SWRL rules are jointly satisfiable under OWL 2 DL semantics.

Second, LCAnet was evaluated against a set of CQs covering all modules. Each CQ was operationalized as SPARQL queries and successfully executed over the instantiated knowledge graph, demonstrating that the ontology adequately supports intended reasoning and querying tasks.

Third, ontology quality was examined using the OOPS! (Ontology Pitfall Scanner!) tool¹¹ [29] which analyses OWL ontologies against a catalogue of common ontology design pitfalls. No critical pitfalls were identified. The analysis reported three important issues, most of which are attributable to modularization and tool-level evaluation limitations rather than conceptual modeling errors. In particular, P11 (missing domain/range, 9 cases) results from incomplete import resolution by the tool, since all nine properties are correctly defined with their respective domains and ranges in their respective modules. P24 (recursive definition, 1 case) is intentionally retained to support flexible classification of LCA results into flow- and impact-related categories of LCA results, formally expressed as $LCA\ Result \sqcap (\exists\ hasFlowCategory.FlowTag) \sqcup (LCA\ Result \sqcap \exists\ hasImpactCategory.ImpactCategory)$. P34 (untyped classes, 2 cases) originates from SWRL rule constructs and does not reflect ontology modeling errors. Additional minor pitfalls include P04 (10 unconnected elements), P08 (60 missing annotations, affecting documentation only), and P13 (24 missing inverse properties), which were intentionally omitted to preserve modularity and design simplicity.

6.2. Reusability and Extensibility

LCAnet was explicitly designed as a modular and extensible ontology network to facilitate reuse across different LCA-related sectors. This design ensures that new impact assessment methods, actors, or activity types can be integrated with minimal disruption to existing components.

A key indicator of reusability is the independent adoption of LCAnet components beyond the original development context. Notably, the Impact Assessment module has been reused in the Cirpass-2 European project for the development of Digital Product Passport ontologies¹², in which it has been extended to serve as the LCA module of the EUDPP Core ontologies, demonstrating external validation and cross-project applicability. Additionally, LCAnet supports any LCA-related study beyond the initial StorAlge prototype use case, such as battery systems which have been used for early simple proof of applicability.

6.3. FAIRness and Design Quality

LCAnet follows FAIR principles through both its publication infrastructure and ontology design. **Findability** is ensured through persistent dereferenceable URIs under w3id.org/lcanet. **Accessibility** is provided via WIDOCO-generated documentation with integrated WebVOWL visualizations and open availability on the project GitHub repository. **Interoperability** is achieved through reuse of established vocabularies (QUDT, TIME, DC Terms, FOAF, Schema.org, GeoNames, VANN), compliance with OWL 2 DL, and SWRL rule integration. **Reusability** is supported by modular decomposition, explicit documentation, competency question coverage, and published SPARQL query patterns, with both human-readable and machine-readable ontology descriptions available.

¹¹<https://oops.linkeddata.es/>

¹²<https://dpp.vocabulary-hub.eu/specifications>

6.4. Adoption, Impact, and Deployment Evidence

LCAnet demonstrates early adoption beyond its development context. The independent reuse of the Impact Assessment module within the Cirpass-2 EU initiative confirms applicability beyond LCA modeling and into broader sustainability data infrastructures, including Digital Product Passport ontology development. From a societal perspective, LCAnet addresses critical needs in environmental impact assessment and sustainable product lifecycle tracking, contributing a reusable ontology network that bridges LCA domain modeling with standardized Semantic Web frameworks.

7. Conclusion

We presented LCAnet, a modular OWL 2 DL ontology network for Life Cycle Assessment structured into six interoperable modules providing a semantic foundation for interoperable LCA knowledge modeling. LCAnet covers all aspects of LCA modeling—core components (activities, flows, and actors), contextual dimensions (spatial and temporal), and descriptive layers (properties, impact characterization, and study metadata)—within a single cohesive framework. It addresses three critical gaps identified in the state of the art: the absence of impact assessment modeling, the lack of semantic richness through expressive axiomatization and rule-based inference, and the absence of modular architecture in existing LCA ontologies.

The ontology was designed through a requirements-driven process grounded in domain expert consultations, formalized as a set of design requirements and competency questions, and evaluated against 20 competency questions verified through SPARQL queries over a populated knowledge graph of 278 individuals. Logical consistency was confirmed using HermiT and Pellet. Applicability was demonstrated on the StorAlge integrated circuit manufacturing use case, with an additional instantiation in a battery manufacturing scenario. Evidence of community reuse is provided by the independent adoption of LCAnet’s Impact Assessment module within the Cirpass-2 EU project for Digital Product Passport ontology development.

LCAnet is publicly available with persistent URIs under <https://w3id.org/lcanet>, with full WIDOCO-generated documentation accessible per module, and all use case data and SPARQL queries available in the project repository.

Several directions remain open for future work. First, LCAnet currently covers the structural and descriptive aspects of LCA modeling but does not model uncertainty in inventory data or impact results—a known challenge in LCA practice. Alignment with provenance ontologies such as PROV-O [30] represents a natural extension in this direction. Second, given the complementary relationship between LCA and Circular Economy, alignment between LCAnet and the CEON ontology network [21] represents a promising direction toward an integrated semantic framework for sustainability assessment—enabling joint reasoning over LCA and circular economy knowledge graphs. Finally, a natural next step toward practitioner adoption is the realisation of the ontology-driven MBSE framework proposed in [31], which introduces a semantic integration architecture bridging LCAnet and SySON—a SysML v2-based modeling editor tailored for LCA practitioners. In this framework, LCAnet serves as the semantic backbone, enabling automated transformation of graphical LCA models into ontology-compliant RDF graphs and supporting reasoning-driven validation. Applying and evaluating this integration constitutes a primary direction for future work within the SmartACV Carnot project.

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Declaration on Generative AI

The authors used Claude (Anthropic) for: grammar and spelling check, paraphrase and reword. After that, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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